FDTD Report

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In the last couple days, I completed Chapter 3 in its entirety: writing my own version of each the codes and recreating each of the figures. For example, Figure 1 is the recreation of the Total-Field/Scattered-Field (TFSF) plot, the chapter's Figure 3.12; and Figure 2 is the recreation of the chapter's final figure, Figure 3.18, which implements a matched-impedance lossy boundary.

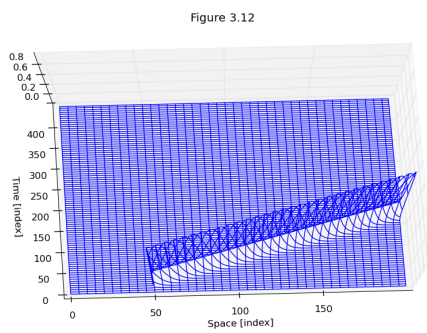

Figure 1: Recreation of Figure 3.12: Waterfall plot of the electric fields produced by Program 1Dtsfsf.c which has a TFSF boundary between nodes hy[49] and ez[50] and simple absorbing boundary conditions.

Once Chapter 3 was finished, I began replacing pieces of the code one-by-one with the more sophisticated C structures and methods described in Chapter 4, sections 1-4. I have not fully completed this step, leaving me currently working on section 4.5.

Future: Along with continuing on in the book, I am working on placing code with the proper (very helpful) color coding within a Latex document. I hope that by next week I will have it figured out so I can easily highlight important parts of the code. Along with that, my figures are not as clean as I would

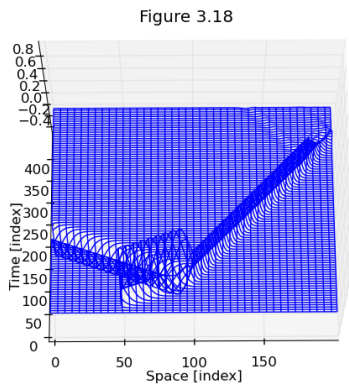

Figure 2: Recreation of Figure 3.18: Waterfall plot of the electric fields produced by 1Dmatched.c which has a dielectric region starting at node 100 with a relative permittivity of 9. This lossless region is followed by a lossy layer with matched impedance. The lossy region starts at node 180.

like, I will get them up to par.

Note on Implementation: To go along with the senior students in the group, I output the data to text files and use python scripts for plotting.

Attatched: In this folder are each of my codes along with recreations of each of the figures in the chapter. The names of the codes are the same as the corresponding codes in the chapter. Also, I included the python script that creates each of the plots.